

ANSWER KEY – Mid Semester Examination 1

Q1. How are continuous traits different from discrete traits? (2 marks)

Expected Answer:

- Discrete traits (qualitative traits): Controlled by one or few genes; phenotypes fall into distinct categories; little environmental influence; Mendelian inheritance patterns (e.g., wrinkled vs smooth peas), one-to-one genotype phenotype maps.
- Continuous traits (quantitative traits): Controlled by many genes (polygenic); phenotypes form a continuous distribution; strong environmental influence; measured in numbers (e.g., height, weight, bristle count), complex genotype phenotype maps.

Q2. Mouse cross – yellow coat, short tail (2 marks)

Answer:

- Both parents: Qq Rr (QQ or RR lethal).
- Viable progeny ratios:
 - Coat: 2 Qq (yellow) : 1 qq (grey)
 - Tail: 2 Rr (short) : 1 rr (normal)
- Independent assortment → Grey + normal = $(1/3) \times (1/3) = 1/9$.

Q3. Heritability and variance components (2 marks)

Given:

- $H^2 = 0.5$, mean = 126, SD = 6.
- $VP = 6^2 = 36$.
- $VG = H^2 \times VP = 0.5 \times 36 = 18$.
- $VE = VP - VG = 18$.

Q4. Dihybrid cross, 9:3:4 F2 ratio → test cross (3 marks)

Answer:

- 9:3:4 = recessive epistasis (e.g., aa epistatic to B/b).
- F1 = Aa Bb (Black).
- Test cross: Aa Bb × aa bb.
- Offspring:
 - Aa Bb = Black
 - Aa bb = Brown
 - aa Bb + aa bb = Golden
- Ratio = 1 Black : 1 Brown : 2 Golden.

Q5. "In X linked loci, the alleles are neither dominant nor recessive. They are just X linked."
– Your opinion? (2 marks)

Expected Nuanced Answer:

The concept of dominance/recessivity applies to diploid individuals, and females are diploid for X-linked genes. However, the expression pattern of X-linked traits depends on the organism:

- In mammals (including humans): X-inactivation (dosage compensation) means females are functional mosaics. In a heterozygote, some cells express one allele, some the other. Thus, dominance is not absolute; traits may show incomplete dominance or codominance at the cellular level, even if the organismal phenotype appears recessive or dominant.
- In non-mammals (e.g., *Drosophila*): No X-inactivation; clear dominance relationships exist just like autosomes.
- In males: Hemizygous; dominance is irrelevant because only one allele is present.

Thus, X-linked alleles absolutely can be dominant or recessive, but their expression is modified by dosage compensation mechanisms in some species.

Q6. Pedigree Analysis (1 + 2 + 2 marks)

The pedigree shows:

- Affected males pass trait to all daughters (who are affected).
- No male-to-male transmission.
- Affected females pass to ~50% sons and ~50% daughters.
- This is X-linked dominant.

a. Most probable mode of inheritance:

- X-linked dominant.
- Strongest evidence:
 - An affected male (II-2) has all daughters affected, no sons affected.
 - This is impossible in X-linked recessive (daughters would be carriers, not affected) and impossible in autosomal dominant (would affect sons and daughters equally).
 - Affected females (e.g., III-2) pass trait to ~half sons and half daughters, consistent with X-linked dominant.

b. V-3 marries normal person – probability first child is affected son?

- V-3 is an affected male and hence ($X^D Y$).

- Affected male ($X^D Y$) cannot pass on the trait to son. The female he is marrying is normal, therefore, ($X^d X^d$).

Therefore, the probability of having an affected son is zero.

Q7. Trihybrid test cross – linkage mapping (2 + 2 + 3 marks)

a. Parental, Single, Double crossovers (2 marks):

- Parental types (most abundant):

Hairy, Red, Tall (1,797) and
Smooth, Blue, Short (1,752).

- Double crossovers (least abundant):

Hairy, Blue, Short (20) and
Smooth, Red, Tall (26).

- Single crossovers: All other classes (to be assigned after gene order).

b. Gene order (2 marks):

Red-Hairy-Tall OR Blue-Smooth-Short (either is fine; order can also be Tall-Hairy-Red OR short-smooth-Blue)

c. Linkage map (3 marks):

RF (Flower–Leaf):

- Recombinants: Hairy+Blue (368+20) + Smooth+Red (26+340) = 388 + 366 = 754.

- RF = $754/4673 = 0.1613 \rightarrow 16.1 \text{ cM}$.

RF (Leaf–Height):

- Recombinants: Hairy + Short (179) + Smooth + Tall (191) + DCOs (20+26) = 179+191+46 = 416.

- RF = $416/4673 = 0.0890 \rightarrow 8.9 \text{ cM}$.

Map:

Flower ——— 16.1 cM ——— Leaf ——— 8.9 cM ——— Height